

MODULE 1

L A N D

Conservation of Natural Resources

Course Code: BCV755B

Terra Study Materials
Department of Civil Engineering



Complete Study Notes

Earth & Crust • Land & Soil • Landforms • Deforestation • Land-Use Change • Soil Health & Degradation • Soil Conservation • Sustainable Land-Use Planning

0. Contents

1	Earth and Formation of Earth's Crust	4
1.1	Formation of Earth	4
1.2	Layers of the Earth — Detailed Explanation	4
1.2.1	1. Crust	4
1.2.2	2. Mantle	5
1.2.3	3. Outer Core	5
1.2.4	4. Inner Core	5
1.3	Formation of Earth's Crust	5
2	Land as a Resource	5
2.1	What is Land?	5
2.2	Soil — Introduction	6
2.2.1	Soil Formation Processes	6
3	Types of Land Forms	7
3.1	Grasslands	7
3.2	Wetlands	7
3.2.1	Functions of Wetlands	7
3.3	Forests	7
3.4	Agricultural Land	7
3.5	Tundra	8
3.6	Deserts	8
3.7	Urban Establishment	8
3.8	Other Landforms	8
4	Conservation of Land Forms	9
5	Deforestation	9
5.1	Important Points about Deforestation	9
5.2	Related Concepts	10
5.2.1	Forest Degradation	10
5.2.2	Forest Fragmentation	10
5.3	Agents vs. Causes of Deforestation	11
5.3.1	Important Agents Leading to Deforestation, Degradation & Fragmentation	11
5.4	Causes of Deforestation	11
5.4.1	1. Direct Causes	11
5.4.2	2. Underlying Causes	12
5.5	Afforestation and Reforestation	12
6	Effect of Land Use Changes	12
6.1	What is Land Use?	12
6.2	Direct vs. Indirect Land Use	13

6.3	Effects of Land Use Changes	13
6.3.1	a. Effects of Land Development	13
6.3.2	b. Effects of Agricultural Land Use	13
7	Soil	15
7.1	Key Points	15
8	Soil Health	15
8.1	Composition of Soil	15
8.2	Indicators of Soil Health	16
9	Ecological and Economic Importance of Soil	16
9.1	Key Points	16
9.2	Benefits of Soil Organic Matter	17
9.3	Importance of Soil — Summary	17
10	Soil / Land Degradation and Its Impact	17
10.1	Causes of Soil / Land Degradation	17
10.1.1	1. Soil Erosion	17
10.1.2	2. Depletion of Soil Fertility	18
10.1.3	3. Landslide Damage	18
10.1.4	4. Soil Sealing through Urbanisation and Industrialisation	18
10.1.5	5. Soil Contamination	19
10.1.6	6. Salination	19
11	Soil Conservation	19
11.1	A Brief History	19
11.2	Strategies to Conserve Soil	20
11.2.1	i. Reduce Soil Disturbance	20
11.2.2	ii. Crop Rotation Practices	20
11.2.3	iii. Cover Cropping Practices	20
11.2.4	iv. Diversify Production Systems	20
11.2.5	v. Add Soil Organic Amendments	20
11.2.6	vi. Integrate Livestock into Cropping Systems	21
11.2.7	vii. Terrace Farming	21
11.2.8	viii. Shelter Belts	21
12	Impact of Soil Degradation on Agriculture and Food Security	21
12.1	Key Points	22
12.2	Sub-Saharan Africa (SSA)	22
12.3	Additional Impact	22
13	Sustainable Land Use Planning	22
13.1	Historical Evolution of the Concept of Land	23
13.2	Sustainable Land Use Helps In	23
13.3	Types of Land Use Planning	23

Summary Table — Module 1: Land25

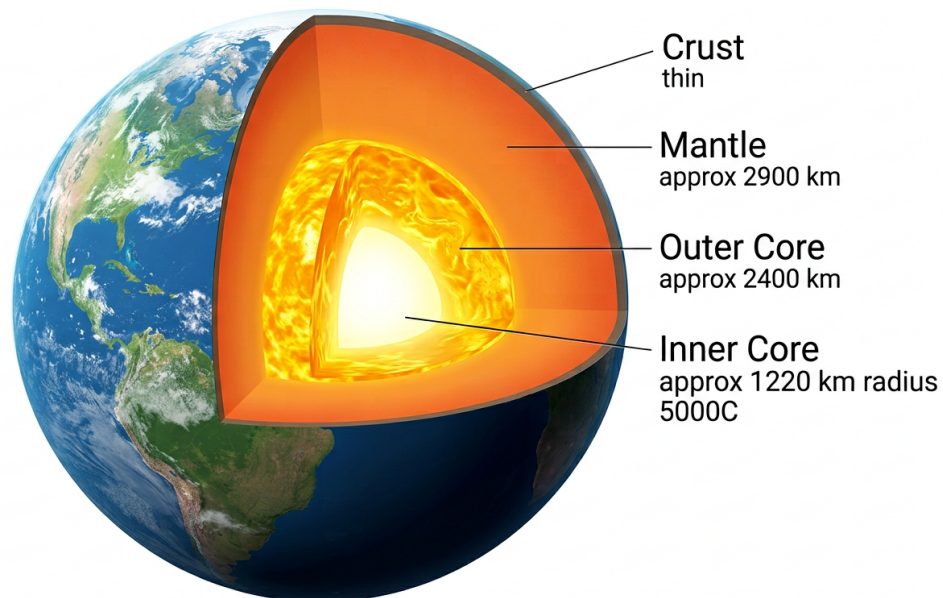
1. Earth and Formation of Earth's Crust

1.1 Formation of Earth

The Earth is thought to be about **4.6 billion years old**. When it first formed, it was not the cool, life-supporting planet we know today — it was a glowing **hot ball of fire and gas**. Over an enormous span of time the outer surface slowly **cooled and compressed into solid rock**, while the material deep inside stayed **molten (liquid)**.

As a result of this cooling-from-outside-inward process, the Earth settled into **four main layers**:

1. **Crust** — the thin, solid outer skin
2. **Mantle** — the thick middle layer
3. **Outer Core** — a liquid metal layer
4. **Inner Core** — a solid metal centre



Cut-away cross-section of the Earth showing its four internal layers.

1.2 Layers of the Earth — Detailed Explanation

1.2.1 1. Crust

- The **outermost solid shell** of the Earth.
- It covers (sits on top of) the underlying **liquid mantle**.
- It is a **very thin layer** — it makes up **less than 1% of Earth's total volume**.
- It forms an important part of the **Lithosphere**.
- The lithosphere is **broken into tectonic plates** that are constantly moving.
- This movement of plates allows **heat to escape** from the Earth's interior out into space.

1.2.2 2. Mantle

- Makes up the **majority of Earth's inner structure**.
- Spans a thickness of about **2900 km**.
- Contains **molten silicate** material.
- The tectonic plates **float on this fluidised mantle**, which keeps them in **continuous motion**.

1.2.3 3. Outer Core

- A **fluid (liquid) layer** roughly **2,400 km thick**.
- Composed mainly of **iron and nickel**.
- Lies **above the inner core** and **below the mantle**.

1.2.4 4. Inner Core

- The **innermost part** of the Earth.
- A **solid core** with a radius of about **1220 km**.
- Its exact composition is **not known**, because no direct sample has ever been taken.
- Its temperature is estimated to be about **5000 °C**.

Remember

Going inward: **Crust (solid, thin) → Mantle (molten silicate, thickest) → Outer Core (liquid iron–nickel) → Inner Core (solid, hottest, ≈5000 °C)**.

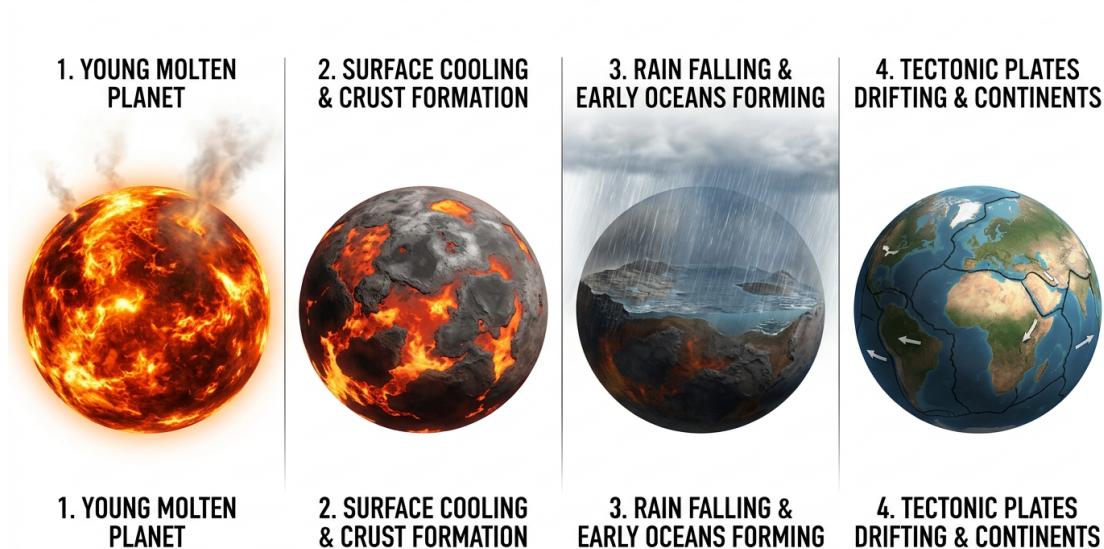
1.3 Formation of Earth's Crust

- About 4.6 billion years ago the hot ball of fire gradually cooled, the surface hardened, and the first **landforms were created**.
- **Atmospheric changes** — especially the precipitation (raining down) of water — **accelerated** this process.
- **Low-lying areas** collected this water and became the **seas and oceans**.
- The hardened **landforms (tectonic plates)** floated on top of the mantle layer.
- The **dynamic (constant) movement** of these tectonic plates slowly shifted the land and produced the **continents** we see today.

2. Land as a Resource

2.1 What is Land?

- Land is a **major constituent** of the life-supporting **Lithosphere**.
- Land makes up about **one-fifth (20%) of the Earth's surface**.
- It is covered by a variety of features: **natural forests, wetlands, grasslands, agricultural land, and urban and rural settlements**.



Stages in the formation of the Earth's crust and continents.

- Land is a **major source of materials** that are essential to humans and to all other living organisms.

2.2 Soil — Introduction

What is Soil?

Soil is the **upper weathered crust of the Earth** that supports plant growth.

- Among all land resources, **soil is of critical importance**.
- Terrestrial (land) plants obtain their **water and mineral nutrients** from the soil.
- When plant and animal materials **decay**, they **release nutrients** back into the soil.
- Soil is a **habitat for many detritus (decomposer) organisms**.
- The study of the origin, formation, and geological distribution of soil is called **Pedology** (*Pedion* = ground, *logos* = discourse).

2.2.1 Soil Formation Processes

Process	Meaning
Weathering	Breaking down of rock into smaller particles
Pedogenesis	Maturation of soil through the development of humus

Tip

Soil takes **several decades or even centuries** to form, because many slow physico-chemical properties are involved in the process.

3. Types of Land Forms

3.1 Grasslands

- One of Earth's **major biomes**.
- The native vegetation of **up to 40% of Earth's land surface**.
- Found on **every continent except Antarctica**.
- **Ecologically and economically important**.
- Provide critical **ecosystem goods and services** at local, regional, and global scales.
- Occur under a **broad range of climates** — but water is generally the limiting factor.
- Many grasslands face **periodic droughts** and have a dormant season, making them **sensitive to climate variability and climate change**.

3.2 Wetlands

- A **distinct ecosystem** that is **flooded by water** — either permanently or seasonally.
- **Oxygen-free (anaerobic) processes** prevail here.
- Their main distinguishing feature is the **characteristic vegetation of aquatic plants** that are adapted to the unique **hydric (waterlogged) soil**.

3.2.1 Functions of Wetlands

- Water **purification** and water **storage**.
- Processing of **carbon and other nutrients**.
- **Stabilisation of shorelines**.
- Support of **plants and animals** — considered the **most biologically diverse** of all ecosystems, home to a wide range of life.

3.3 Forests

- A **complex ecosystem** made up mostly of **trees and shrubs** with a **closed canopy**.
- Storehouses of a large variety of **life forms** — plants, mammals, birds, insects, reptiles.
- Contain **abundant microorganisms and fungi** that decompose dead organic matter and **enrich the soil**.
- Nearly **4 billion hectares** of forest cover the Earth's surface.
- Roughly **30% of Earth's total land area** is forested.

3.4 Agricultural Land

- Land **devoted to agriculture** — the systematic and controlled use of land for:
 - **Rearing livestock**, and
 - **Producing crops** to make food for humans.
- It is synonymous with **farmland, cropland, pasture, or rangeland**.

3.5 Tundra

- A biome where **tree growth is hindered** by low temperatures and short growing seasons.
- Vegetation: **dwarf shrubs, sedges, grasses, mosses, and lichens.**
- Tundra soil is **rich in nitrogen and phosphorus.**
- Three types: **Arctic tundra, Alpine tundra, Antarctic tundra.**

3.6 Deserts

- A **barren area** of landscape where **very little precipitation occurs.**
- Living conditions are **hostile** for plant and animal life.
- Can be classified by: amount of **precipitation, temperature, causes of desertification,** and **geographical location.**
- Formed by **weathering** — the large temperature difference between day and night strains the rocks until they break into pieces.
- Although rain is rare, occasional **downpours can cause flash floods.**

3.7 Urban Establishment

- Land used for **buildings, roads, and infrastructure** for human settlements.
- Represents a **human-modified environment.**



A visual comparison of the seven major landform types.

3.8 Other Landforms

In addition to the major types above, the following landforms also exist: **Mountain Range, Glacier, Volcano, Plateau, Canyon, Valley, Lake, River, Delta, Basin, Desert, Inlet, Strait, Peninsula, Archipelago, and Island.**

4. Conservation of Land Forms

Landform Conservation

Landform conservation is the **protection and wise use of the land base** — including its form, its soil, and the associated biophysical (natural living and non-living) processes.

It is an approach that encourages **planning, design, and construction practices** which:

1. **Minimise disruption** to the natural form and related ecological processes.
2. **Enhance the protection** of biophysical features in their natural state.
3. Keep a greater portion of a site in an **open-space character**.
4. **Protect endangered flora and fauna** from extinction.
5. **Preserve traditional practices** associated with these areas (which are of great cultural importance).
6. Prevent **drastic effects on climate** — because changes to a landform can cause global climatic concerns.

5. Deforestation

Deforestation

Deforestation is the **conversion of forested areas to non-forest land use** — such as arable land, urban use, logged area, or wasteland.

FAO Definition

The **Food and Agriculture Organisation (FAO)** of the UNO defines deforestation as: *“The conversion of forest to another land use **OR** the long-term reduction of tree canopy cover **below the 10% threshold.**”*

5.1 Important Points about Deforestation

- It can be **deliberate** (done on purpose for agriculture or urban development).
- It can be **unintentional** (for example, a consequence of uncontrolled grazing, which prevents the natural regeneration of young trees).
- The combined effect of **grazing + fires** is a major cause of deforestation in dry areas.
- It implies a **long-term (>10 years) or permanent loss** of forest cover.
- It is **not just the conversion** of forest to non-forest, but also **degradation** that reduces:
 - Forest quality,
 - Density and structure of trees,
 - Ecological services,
 - Biomass of plants and animals,

- Species diversity, and
- Genetic diversity.

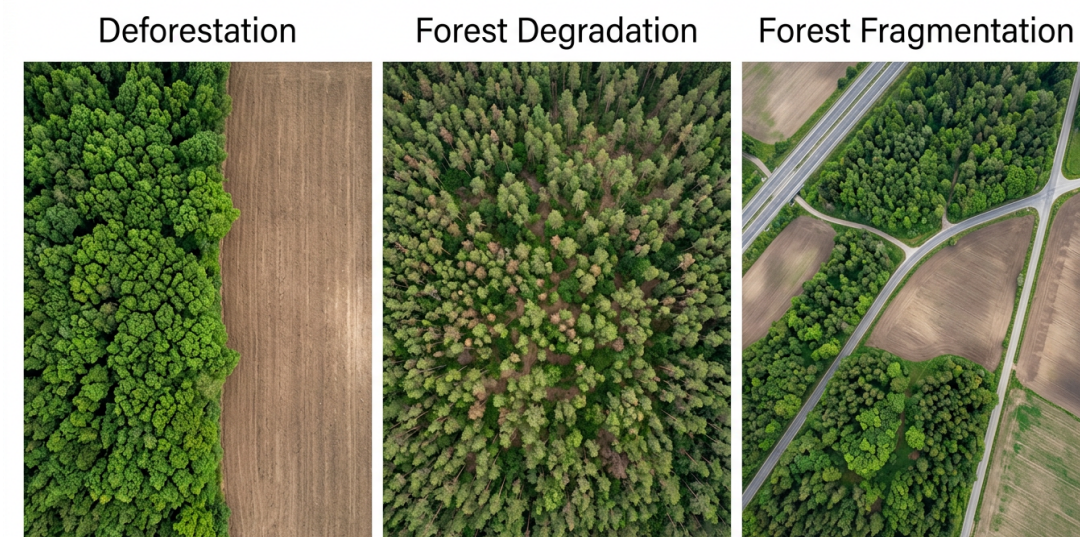
5.2 Related Concepts

5.2.1 Forest Degradation

- A process leading to a **temporary or permanent deterioration in the density or structure of vegetation cover or species composition**.
- It is a change in forest attributes that leads to a **lower productive capacity**, caused by increased disturbances.
- It is different from deforestation, but the two terms are often used interchangeably.

5.2.2 Forest Fragmentation

- Any process that **breaks a formerly continuous forest into patches** separated by non-forested land.
- A **complex phenomenon** resulting from the dynamic interaction between the natural landscape and society's ever-increasing demand for land.
- It creates a **mosaic** of natural and human-modified environments.



Deforestation, forest degradation and forest fragmentation compared.

5.3 Agents vs. Causes of Deforestation

Agents	Causes
Those who clear the forest — individuals, corporations, government agencies, development projects	The processes that lead to deforestation

5.3.1 Important Agents Leading to Deforestation, Degradation & Fragmentation

Agent	Link to Deforestation
Slash-and-burn farmers	Clear forest to grow subsistence and cash crops
Commercial farmers	Clear forest to plant commercial cash crops
Cattle ranchers	Clear forest to plant pasture
Livestock herders	Intensification of herding activities
Loggers	Remove commercial timber; their roads give access to other users
Commercial tree planters	Clear forest fallow to establish plantations
Firewood collectors	Intensification of collection can lead to deforestation
Mining & petroleum industries	Roads and seismic lines give access; localised deforestation
Land settlement planners	Relocation of people into forested areas
Infrastructure developers	Road/highway construction through forests; flooding by dams

5.4 Causes of Deforestation

Deforestation is attributed to **socio-demographic factors** such as population growth and the political economy.

5.4.1 1. Direct Causes

- **Logging** — loggers build roads into primary forests and remove trees; logging destroys 45–74% of the residual (remaining) trees.
- Conversion of forested land for **agriculture**.
- **Cattle-raising**.
- **Urbanisation**.

- Mining and oil exploitation.
- Acid rain.
- Forest fire.

Historical Note

Countries like **Ireland and Scotland** were once almost entirely forested, but were nearly completely cleared under British rule to provide timber for English shipbuilders.

5.4.2 2. Underlying Causes

- Unpriced forest goods and services
- Monopolies in the market
- Wrong incentives
- Improper regulatory measures
- Illegal activities in forest land
- Weak ownership by the government
- Population growth
- Economic growth
- Excessive consumption
- Global warming
- War

5.5 Afforestation and Reforestation

Afforestation

The **conversion from other land uses into forest**, or an increase of canopy cover **above the 10% threshold**. It is the **reverse of deforestation**. It includes areas actively converted through **silvicultural measures**, and also **natural transitions** into forest (e.g. abandoned agricultural land or burnt-over areas).

Reforestation

The **re-establishment of forests** after deforestation, **due to natural sources**.

6. Effect of Land Use Changes

6.1 What is Land Use?

- “**Land use**” refers specifically to the **purposes for which land is occupied or managed** for human ends.
- Common categories: **cropland, pasture, and settlements**.
- Land **not used by humans** is designated as “**wild**” or “**natural**” land.
- Land **lightly used** by humans is designated as “**semi-natural**”.

6.2 Direct vs. Indirect Land Use

Type	Definition	Example
Direct Land Use	A specific piece of land and how it is being used at a specific moment	Cattle rearing in pasture land — cows grazing directly
Indirect Land Use	The “use” happens later in the value chain, when a product is consumed or used	Cattle rearing in cropland — converting wild land into fodder cropland

6.3 Effects of Land Use Changes

Land use changes occur **constantly and at many scales**. They can have **specific and cumulative effects** on: air and water quality, watershed function, generation of waste, extent and quality of wildlife habitat, climate, and human health.

6.3.1 a. Effects of Land Development

Land development creates **impervious surfaces** (surfaces water cannot pass through) through roads, parking lots, and other structures.

Impacts of impervious surfaces:

- Contribute to **nonpoint-source water pollution** by limiting the soil’s capacity to filter runoff.
- Affect **peak flow and water volume**, which heightens erosion potential.
- **Increase storm-water runoff**, delivering more pollutants (dirt, oils, fertiliser nutrients, toxic compounds) to water bodies.
- **Affect groundwater aquifer recharge**.

Point-source discharges: Industrial and municipal wastewater facilities contribute **toxic compounds and heated water**.

Suburbanisation effects:

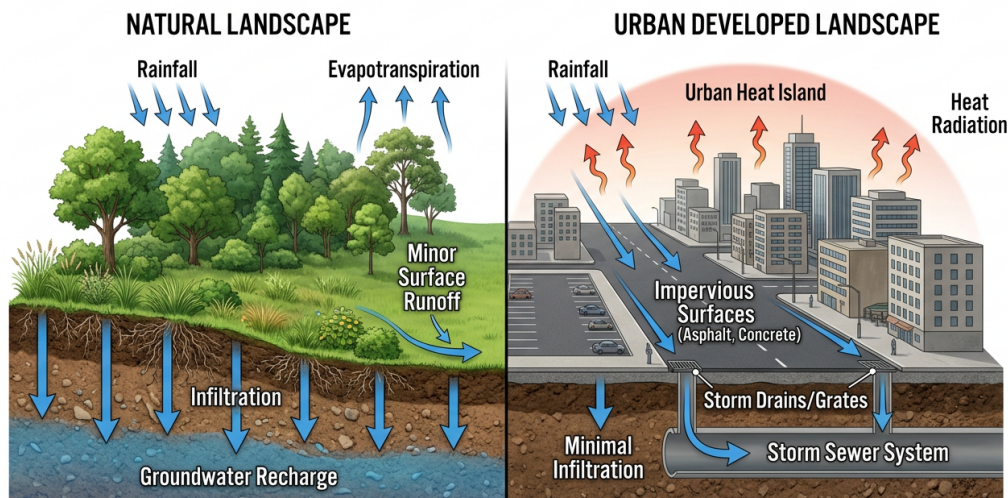
- **Increased air pollution** from vehicle use raises air-pollutant concentrations and worsens health problems like **asthma**.
- Formation of **“heat islands”** — domes of warmer air over urban areas, caused by the loss of trees/shrubs and the absorption of heat by pavement and buildings.
- Heat islands affect **local, regional, and global climate**, as well as air quality.

6.3.2 b. Effects of Agricultural Land Use

Agricultural land uses affect the **quality of water and watersheds**:

- The types of crops planted, the tillage practices, and the irrigation used can **limit the water available** for other uses.
- **Livestock grazing in riparian (riverbank) zones** reduces stream-bank vegetation, which increases water temperature, sedimentation, and nutrient levels.
- **Runoff from pesticides, fertilisers, and animal manure** degrades water quality.

COMPARISON: NATURAL VS. URBAN LANDSCAPE (WATER INFILTRATION & HEAT ISLAND EFFECT)



How impervious urban surfaces and “heat islands” affect the environment.

- It may cause a **loss of native habitats**, or increased **wind erosion and dust** that exposes humans to particulate matter and chemicals.
- Some land uses **accelerate the spread of invasive species**.
- Agricultural practices (overgrazing, land conversion, fertilisation, use of chemicals) enhance the growth of **invasive plants**, which alters fish and wildlife habitat, reduces biodiversity, and creates health risks.

Shire River, Malawi, Africa

The **Shire River** in southern Africa is a prime example of where **Land Use Land Cover Change (LUCC)** has caused severe soil erosion and heavy siltation downstream. The river is important for **agriculture, domestic purposes, and electricity generation**. The **Nkula B Hydroelectric Power Station** is situated across the Shire River, and its dam has been threatened by **massive siltation** caused by increased human population and agricultural activity.

Key findings:

- A **strong link exists** between land use change and soil erosion.
- Good vegetation cover **promotes infiltration** and soil retention.
- **Deforestation** causes increased runoff and stronger soil erosion.
- Erosion of soil under continuous cultivation is the **most serious form of resource degradation in Malawi**.
- Rate of soil loss: currently **29 t/ha/year** (higher than the previous estimate of 20 t/ha/year).

7. Soil

Soil — SSSA Definition

The **Soil Science Society of America (SSSA)** defines soil as: “*The unconsolidated mineral or organic material on the immediate surface of the earth that serves as a **natural medium for the growth of land plants.***”

7.1 Key Points

- Soil is composed of: **inorganic particles, organic matter, air, water, and a variety of organisms.**
- It takes **decades or even centuries** to develop soil horizons (layers).
- Human activities create worldwide problems such as **soil erosion** and the **depletion of fertility.**
- The **yield of all biotic (living) products** in terrestrial ecosystems depends on **soil fertility.**

8. Soil Health

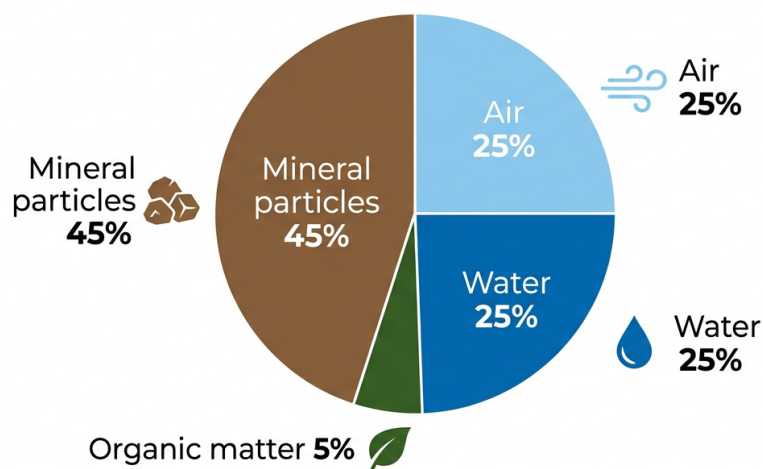
Soil Health

The state of the soil being in **sound physical, chemical, and biological condition**, having the capability to sustain the **growth and development of land plants.**

8.1 Composition of Soil

Component	Percentage
Soil mineral particles	45%
Air	25%
Water	25%
Organic matter	5%

Composition of Soil



Composition of a healthy soil shown as a pie chart.

8.2 Indicators of Soil Health

Physical Indicators	Chemical Indicators	Biological Indicators
Soil compaction level	Primary nutrients (N, P, K)	Soil organic matter
Soil bulk density	Micronutrients (Zn and Fe)	Permanganate oxidizable carbon
Dry aggregate size distribution	Soil salinity	Nitrogen mineralization
Wet aggregate stability	Soil sodicity (sodium problem)	Carbon mineralization
Soil texture	Soil pH	Total microbial biomass
Available water capacity	—	Microbial diversity index

9. Ecological and Economic Importance of Soil

9.1 Key Points

- Soil is the **foundation of basic ecosystem function**.
- Soil **filters water** and provides essential nutrients to forests and crops.
- It helps **regulate the Earth's temperature** and the important greenhouse gases.
- Almost **35% of all greenhouse gases (GHGs)** released due to human (anthropogenic) activities since 1850 are linked to **land-use changes**.
- **Soil stores CO₂** and other GHGs in soil organic matter — this is called **carbon sequestration**.

- Crop, grazing, forest lands, and wetlands all have the potential to **mitigate GHG emissions** through proper management.

9.2 Benefits of Soil Organic Matter

- Filters and **cleans water**
- Enhances **water retention and storage**
- **Mitigates the impact** of extreme weather events
- Improves **soil structure**
- **Reduces soil erosion**
- Provides **microbial habitats**
- Source of **long-term, slow-release nutrients**

9.3 Importance of Soil — Summary

Function	Description
Food and biomass production	Supports all agricultural production
Environmental interaction	Storage, filtering, and transformation of substances
Biological habitat & gene pool	Home to countless organisms
Source of raw materials	Clay, minerals, construction materials
Physical and cultural heritage	Connection to land heritage
Platform for man-made structures	Buildings, highways

10. Soil / Land Degradation and Its Impact

10.1 Causes of Soil / Land Degradation

1. Soil Erosion
2. Loss of Soil Fertility
3. Landslide
4. Soil sealing through urbanisation and industrialisation
5. Soil contamination
6. Salination

10.1.1 1. Soil Erosion

- The **movement of water and air removes the topsoil** from the land.
- Abundant plant cover **significantly reduces** soil erosion.
- **Human activities accelerate** erosion by removing the natural plant cover.

- From croplands in India, **millions of tons of topsoil** are eroded into the sea each year.
- Erosion causes a **significant loss of soil fertility**, because it transports organic matter and nutrients away.
- Eroded soil enters **streams, rivers, and lakes as sediment**, affecting water quality and the habitats of aquatic organisms.

10.1.2 2. Depletion of Soil Fertility

- When natural vegetation is removed to develop agricultural systems:
 - The **nutrients stored in the vegetation are removed**, and
 - The **organic matter and nutrients accumulated in the soil are also lost**.
- From agricultural systems, **nutrients are exported through the crop harvest**.
- Over time, agricultural soil **inevitably loses its fertility**.

10.1.3 3. Landslide Damage

- A landslide refers to all kinds of **rapid mass movement** — involving rapid flow, slip, falling, sliding, or flowing (or a combination).
- Since it moves **soil and vegetation mass**, it causes:
 - Fragmentation of the self-sustaining **soil–plant system**,
 - **Disruption of community structure**,
 - Disruption of **nutrient cycling**,
 - **Loss of soil with nutrients** from the ecosystem,
 - **Destruction of habitat** and loss of biodiversity,
 - **Decrease in forest biomass and soil fertility**.
- Recovery depends on the **intensity of the disturbance**: low intensity → **faster recovery**; long-term disturbance → **delayed recovery**.

10.1.4 4. Soil Sealing through Urbanisation and Industrialisation

- **Urbanisation** = establishing infrastructure for modern life (housing, roads, land developments).
- **Soil Sealing** = the closure of soil pores on the surface, due to the materials used in urbanisation.

Effects of soil sealing:

- Soil becomes **unable to absorb rainwater** for infiltration.
- It cannot perform its **general filtering** functions.
- It **changes water-flow patterns** in the surrounding soils.
- Soil sealing is **almost irreversible**.
- There is growing concern among governments and environmental regulators about this **permanent loss of soil** and of ecosystem functions.

10.1.5 5. Soil Contamination

- The **introduction of contaminants into the soil**, causing damage or loss of soil functions.
- Can lead to the **contamination of water**.
- Contaminants above certain levels cause **negative consequences** for the **food chain**, **human health**, **all ecosystems**, and other natural resources.

10.1.6 6. Salination

- The **accumulation of soluble salts** (mainly sodium, magnesium, and calcium) in the soil.
- To the extent that **crop production is severely reduced**.
- Often associated with **insufficient irrigation practices**:
 - Irrigation water contains variable amounts of salts.
 - In regions with **low rainfall** and **high evapotranspiration**, salts build up in the surface layer.
 - The salts cannot be washed out, so they **build up in the soil surface layer**.

CAUSES OF SOIL DEGRADATION



The six major causes of land / soil degradation.

11. Soil Conservation

11.1 A Brief History

- **1907** is an important year in the history of soil conservation.
- The **United States Department of Agriculture** declared an official policy of **land protection**.
- **Iceland** established the **world's first soil conservation service** in the same year.

11.2 Strategies to Conserve Soil

11.2.1 i. Reduce Soil Disturbance

- **Intensive soil tillage** is one of the greatest catalysts of soil degradation.
- **Plough tillage** systems are the **most detrimental practice** to soil health (both long- and short-term).
- Reduced-tillage methods include **strip tillage** and **no tillage**.
- These can produce **similar crop yields** to conventional plough-till methods in arid and semi-arid farmlands.

11.2.2 ii. Crop Rotation Practices

- Farmers grow crops from **different plant families** in a well-planned sequence over **multiple seasons** in the same field.
- It can be of **any length**, depending on the farmer's preference or local regulations.
- Key factor: a **variety of crops from different families** grown in the same soil.
- **Benefits:** breaks the **disease cycle** in a field; greater crop diversity gives greater **diversity of soil microbiomes**, leading to better soil health.

11.2.3 iii. Cover Cropping Practices

- Cover crops are plants grown **in between cash-crop cycles** to **protect land from erosion** and add **root exudates and biomass** for soil improvement.
- The cover-crop biomass **must be returned to the soil** after the growing period for full benefit.
- **Legumes fix nitrogen** in the soil through an association with bacteria called **Rhizobium**:
 - Rhizobium converts **atmospheric nitrogen into plant-useable nitrogen**.
 - When leguminous cover crops are incorporated into the soil, that nitrogen becomes available for the **next crop**.
 - This **reduces the external nitrogen demand** of the following cash crop.

11.2.4 iv. Diversify Production Systems

Diversified farming includes mixed varieties and **mixed cropping**, **integrating livestock** into crop production, **cover cropping**, **crop rotation**, **fallowing fields**, and **hedgerows and buffer strips**.

- **Benefits:** improvements in **nutrient cycling**, better **soil water management**, better **pest control**, and habitats for **pollinators**.

11.2.5 v. Add Soil Organic Amendments

- Organic amendments are any material of **plant or animal origin** added to the soil to **improve soil conditions** and **stimulate biodiversity**.
- Examples: **manure**, **compost**, **biochar**, and similar materials.
- Regular addition improves **soil fertility**, **biological functions**, and **physical characteristics**.

11.2.6 vi. Integrate Livestock into Cropping Systems

- Integrating livestock through **winter grazing** increases the **efficiency of farm-resource use**, decreases the need for **synthetic fertilisers and pesticides**, and improves **soil health**.
- It supports **diverse microbial communities**, promoting soil organic-matter accumulation.
- **Grazing speeds up decomposition** of leftover plant residues, enhancing the availability of **nitrogen, phosphorus, and sulphur**.
- The **hoof action** of animals brings more residue in contact with the soil, and the **saliva** of grazing animals helps the growth and release of nutrients.

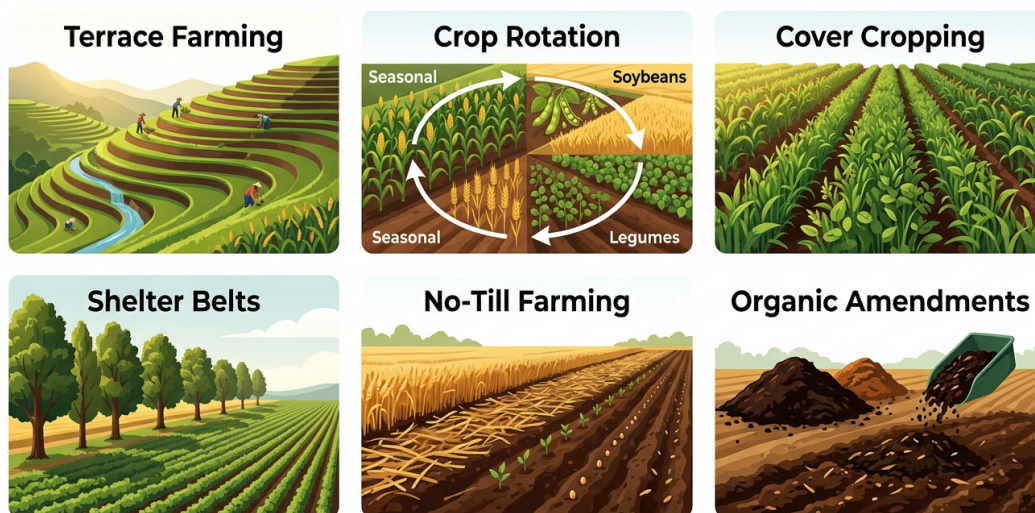
11.2.7 vii. Terrace Farming

- The practice of **cutting flat areas out of a hilly or mountainous landscape** to grow crops.
- Practised from the **rice fields of Asia** to the **steep slopes of the Andes** in South America.
- **Benefits:** helps **prevent erosion and soil runoff**; a hillside can remain **productive for a long time** as long as the soil is properly cared for.

11.2.8 viii. Shelter Belts

- In **coastal and dry regions**, rows of trees are planted to **check wind movement** and protect the soil cover.
- They act as a natural **wind barrier** to prevent wind erosion.

Soil Conservation Techniques



Key soil conservation techniques used by farmers.

12. Impact of Soil Degradation on Agriculture and Food Security

12.1 Key Points

- The gravity of soil degradation depends on the **type of degradation process**.
- **Soil erosion and salinification** are very serious — they can drive farmers to **abandon the land**, or to face **very high management costs**.

Global Perspective — India

India supports **18% of the world's human population** and **15% of the world's livestock population**, but has only **2.4% of the world's land area**. Soil degradation here is causing a **decline in crop productivity**, **huge economic loss**, and a **threat to food security** and to the livelihoods of farmers.

12.2 Sub-Saharan Africa (SSA)

- **Nutrient depletion** is the primary form of soil degradation in SSA.
- It leads to a **decline in crop productivity**.
- It is linked to **hunger and poverty**.

12.3 Additional Impact

- Soil degradation forces farmers to look for **new land**.
- Most new land is **marginal land** or **tropical forest** (especially in Latin America and Africa).
- Such soils are **not suitable for agricultural production** and require **high investment** to become productive.
- The overall effect of soil degradation is a **major threat to food security**, especially in poor regions.

13. Sustainable Land Use Planning

Land Use Planning

The **systematic assessment** of land and water potential, of alternatives for land use, and of economic and social conditions — in order to **select and adopt the best land-use options**.

Purpose: to select and put into practice the land uses that will **best meet the needs of people** while **safeguarding resources for the future**.

13.1 Historical Evolution of the Concept of Land

Time Period	Perception of Land
1700s	Land = Wealth
Later	Land = Commodity
Further later	Land = Scarce Resource
From the 1980s onward	Land = Scarce Community Resource (both commodity and wealth)

Sustainable Land Use

The use of components of land in a way, and at a rate, that **does not lead to the long-term decline of land** — thereby maintaining its potential to **meet the needs and aspirations of present and future generations**.

Sustainable Land Management (SLM) covers the **ecological, economic, and socio-cultural** dimensions of sustainable development.

13.2 Sustainable Land Use Helps In

1. Protecting agricultural land from urban and peri-urban encroachment.
2. Protecting natural capital from urban and peri-urban encroachment.
3. Preventing or limiting exposure of saline and acid-sulphate soils.
4. Rehabilitation and/or avoidance of contaminated sites.
5. Adapting to salinisation and rising groundwater levels.
6. Ensuring land use reflects land capability.
7. Protecting groundwater quality and quantity.
8. Protecting water quality and minimising erosion through water-sensitive urban design.
9. Minimising eutrophication and other pollution of surface and groundwater.
10. Establishing appropriate buffers between development, coastal estuaries, and water foreshores.
11. Floodplain management.
12. Preventing or limiting vegetation clearing.
13. Protecting natural habitats from destruction and fragmentation.
14. Preserving and enhancing ecological corridors.
15. Reducing car dependence through transport-demand management.
16. Accounting for sea-level rise and increased storm surge from coastal development.

13.3 Types of Land Use Planning

- Spatial Land Use Planning
- Participatory Land Use Planning
- Ecosystem-based Land Use Planning
- Regional Land Use Planning
- Integrated Land Use Planning
- Territorial Ecological Planning
- Urban Land Use Planning

13. Summary Table — Module 1: Land

Topic	Key Concept
Earth Formation	4.6 billion years ago, a hot ball cooled → solid crust formed
Layers of Earth	Crust, Mantle, Outer Core, Inner Core
Land as a Resource	1/5th of Earth's surface; forests, grasslands, wetlands, agricultural land
Soil Formation	Weathering + Pedogenesis
Types of Land	Grassland, Wetland, Forest, Agricultural, Tundra, Desert
Landform Conservation	Protect, minimise disruption, preserve biophysical features
Deforestation	Conversion of forest to non-forest; FAO threshold = below 10% canopy
Forest Degradation	Temporary/permanent deterioration in density/structure of vegetation
Forest Fragmentation	Continuous forest broken into patches by non-forested land
Afforestation	Converting other land to forest (canopy above 10%)
Reforestation	Re-establishment of forests by natural means
Soil Health	Sound physical, chemical, and biological condition of soil
Soil Degradation Causes	Erosion, fertility depletion, landslide, sealing, contamination, salination
Soil Conservation	Reduced tillage, crop rotation, cover crops, terracing, shelter belts
Impact on Food Security	Reduces crop productivity, threatens livelihoods, linked to hunger/poverty
Sustainable Land Use	Meeting present needs without compromising future generations'

In a nutshell

These are the complete, detailed study notes of **Module 1: Land (BCV755B — Conservation of Natural Resources)**, covering every topic in the module — from the formation of the Earth to sustainable land-use planning — without skipping anything.